	AIR UNIVERSITY
	DEPARTMENT OF COMPUTER ENGINEERING
	EXPERIMENT NO: 1

Lab Tittle: Introduction to CISCO Packet Tracer (Hub & Switch)

Name:	Muhammad Waqas	**Reg No:**	240756
	Faiq Ali		249520

Objectives: The purpose of this lab is to become familiar with building topologies in Packet Tracer

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks:_____ Obtained Mark: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks: _____



DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

OPERATING SYSTEMS LABORATORY

Lab Report: 1

Tittle: Introduction to CISCO Packet Tracer (Hub & Switch)

Name: Muhammad Waqas, Faiq Ali

Reg. No: 240756, 249520

Submitted To: CE- Engr. Basharat Ahmad

Date: 18 Sep, 2026

Experiment No: 1
Introduction to CISCO Packet Tracer (Hub & Switch)

Lab Task 1: Write the advantages and disadvantages of different network topologies (Mesh, Star, Bus and ring)?

Mesh Topology

Advantages of Mesh Topology:-

- **High Reliability** Mesh topology eliminates single points of failure. If one node or connection fails, data can be rerouted through alternative paths, ensuring continuous network operation. This redundancy makes it ideal for critical applications like healthcare systems and financial networks.
- **Enhanced Security** Multiple data paths make it difficult for attackers to intercept or compromise network communications. The distributed nature allows implementation of various security protocols, including encryption and authentication, across multiple routes.
- **Fault Isolation** Failed nodes can be quickly identified and isolated without affecting the entire network. This feature is particularly valuable in mission-critical environments where network downtime is unacceptable.
- **High Performance** Direct connections between nodes enable fast data transmission with minimal latency. This makes mesh topology suitable for applications requiring high-speed data transfer, such as real-time communications and multimedia streaming.
- **Load Distribution** Network traffic can be distributed across multiple paths, preventing congestion and optimizing overall network performance.

Disadvantages of Mesh Topology:-

- **High Cost** The major drawback is the extensive cabling requirements. For n nodes, n(n-1)/2 connections are needed, making it expensive to implement and maintain as network size grows.
- **Complexity** Managing numerous connections becomes increasingly difficult as the network expands. Configuration, troubleshooting, and maintenance require specialized expertise and significant time investment.
- **Limited Scalability** Adding new nodes requires connecting them to all existing nodes, creating exponential growth in connections and costs. This makes large-scale implementations impractical.
- **Bandwidth Overhead** Multiple connections can lead to bandwidth wastage and potential congestion, especially when not all paths are optimally utilized.
- **Physical Space Requirements** The extensive cabling infrastructure requires significant physical space and careful cable management, particularly challenging in dense network environments.

Star Topology

Advantages of Star Topology:-

- **High Reliability:** If one cable or device fails, only that single device disconnects while the rest of the network keeps working.
- **Easy Expansion:** You can add or remove devices easily by plugging them into the central hub without disturbing other users.
- **Low Collision Risk:** Dedicated connections for each device reduce data traffic collisions.
- **Simple Troubleshooting:** Faults and broken cables are easy to detect and isolate because each wire runs directly to the central unit.
- **Centralized Management:** Network traffic and security are easy to monitor from the central switch or router

Disadvantages of Star Topology:-

- **Single Point of Failure:** If the central hub or switch breaks down, the entire network stops working.
- **Higher Cost:** It requires more cable length and extra hardware (hubs or switches) compared to a linear bus layout.
- **Cable Dependency:** Long individual cable runs are required for every single device, which increases installation effort and material costs.
- **Maintenance Needs:** The central device requires power and regular upkeep to handle all network traffic effectively

Bus Topology

Advantages of Bus Topology:-

- **Low Cost:**It uses less cable than a star layout, making it cheaper to build.
- **Easy setup:** It is simple to install and connect new devices.
- **Good for small spaces:** It works well for small, temporary, or low-traffic networks.
- **Simple design:** The linear structure is easy to understand and expand on a small scale. [1, 2, 3]
-

Disadvantages of Bus Topology:-

- **Single point of failure:**If the main backbone cable breaks, the whole network stops working.
- **Slow performance with scale:** Adding more devices slows down data transmission.
- **Hard to troubleshoot:** Finding a break in the cable or a broken device is difficult.
- **Traffic limits:** High data traffic causes high rates of packet loss and data collisions.

- **Security risks:** Every connected device sees all transmitted data, which lowers privacy.
- **Extra hardware needed:** Both ends of the main cable require terminators to stop signal bouncing. [1, 2, 3, 4, 5]

Ring Topology

Advantages of Ring Topology:-

- **Low collision risk:** Data moves in one direction using token passing, which prevents multiple devices from sending data at the same time.
- **Equal access:** Every device gets a fair, orderly turn to transmit data.
- **No central hub:** The network operates without needing a central controlling server or hub.
- **Predictable performance:** Traffic flow remains steady and organized even under heavy load

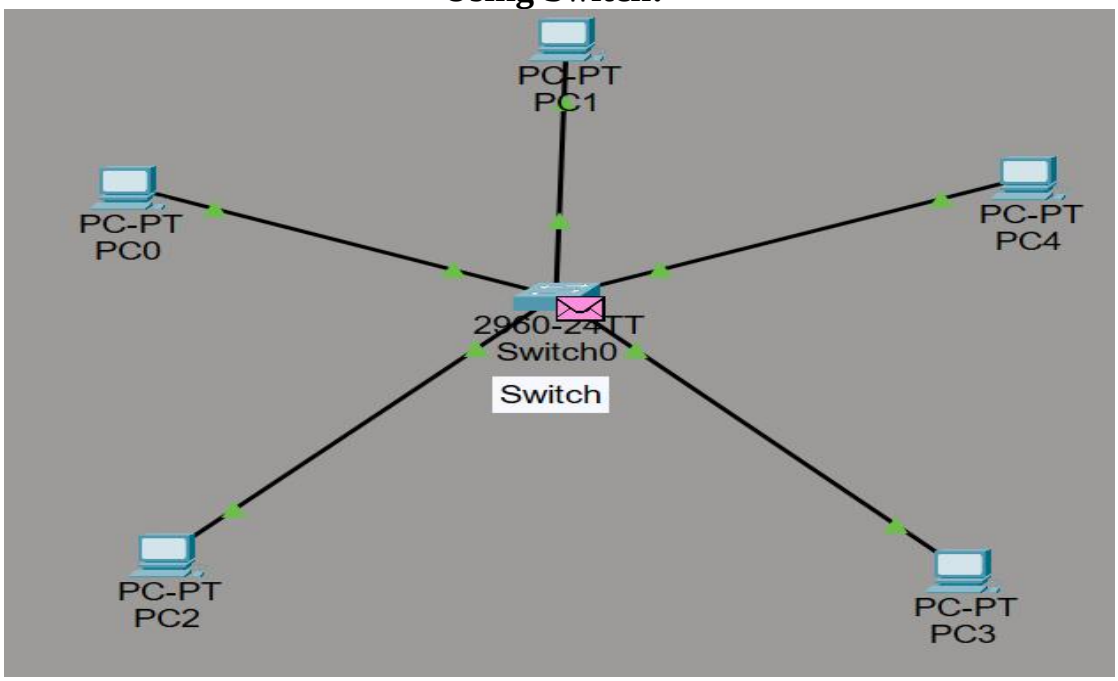
Disadvantages of Ring Topology:-

- **Single point of failure:** If one device or cable breaks in a standard single-ring setup, the whole network goes down.
- **Slow speed:** Data must pass through every intermediate node to reach its destination, adding delay.
- **Hard to expand:** Adding or removing devices usually requires breaking the loop and disrupting network activity.
- **Difficult troubleshooting:** Finding the exact spot of a break or cable fault takes time. [1, 2, 3, 4, 5]

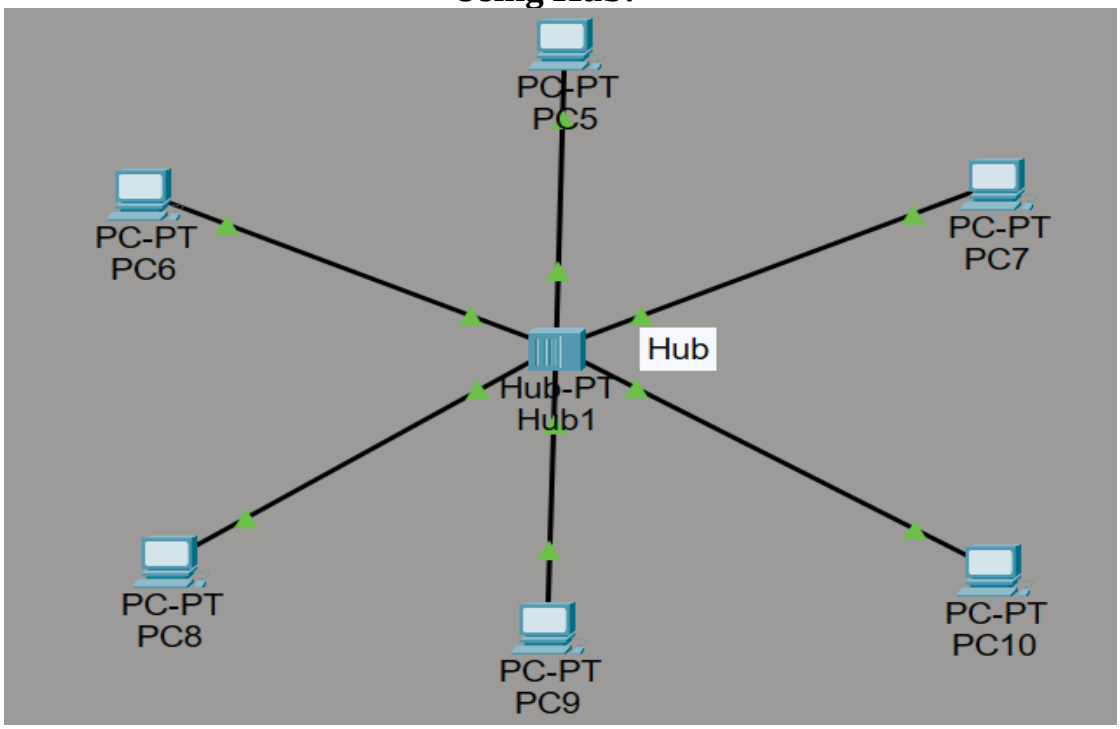
Lab Task 2: Implement the following topologies in packet tracer and attach screen shots ((Mesh, Star, and ring)?

Start Topology

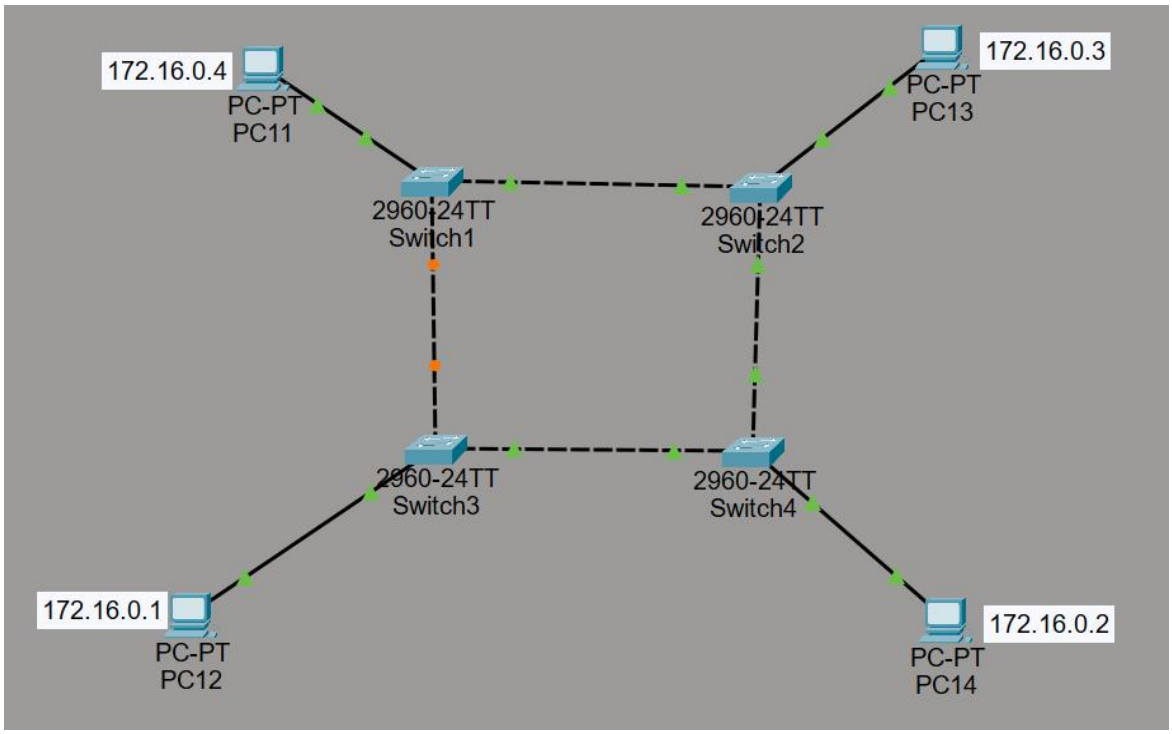
Using Switch:-



Using Hub:-



Ring Topology



Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device
	0.049	--	Switch0
	0.051	Switch0	PC2
	0.051	Switch0	PC4
	0.051	Switch0	PC1
	0.051	Switch0	PC0
	0.051	Switch0	PC3

Reset Simulation | Constant Delay | Captured to: 0.051 s

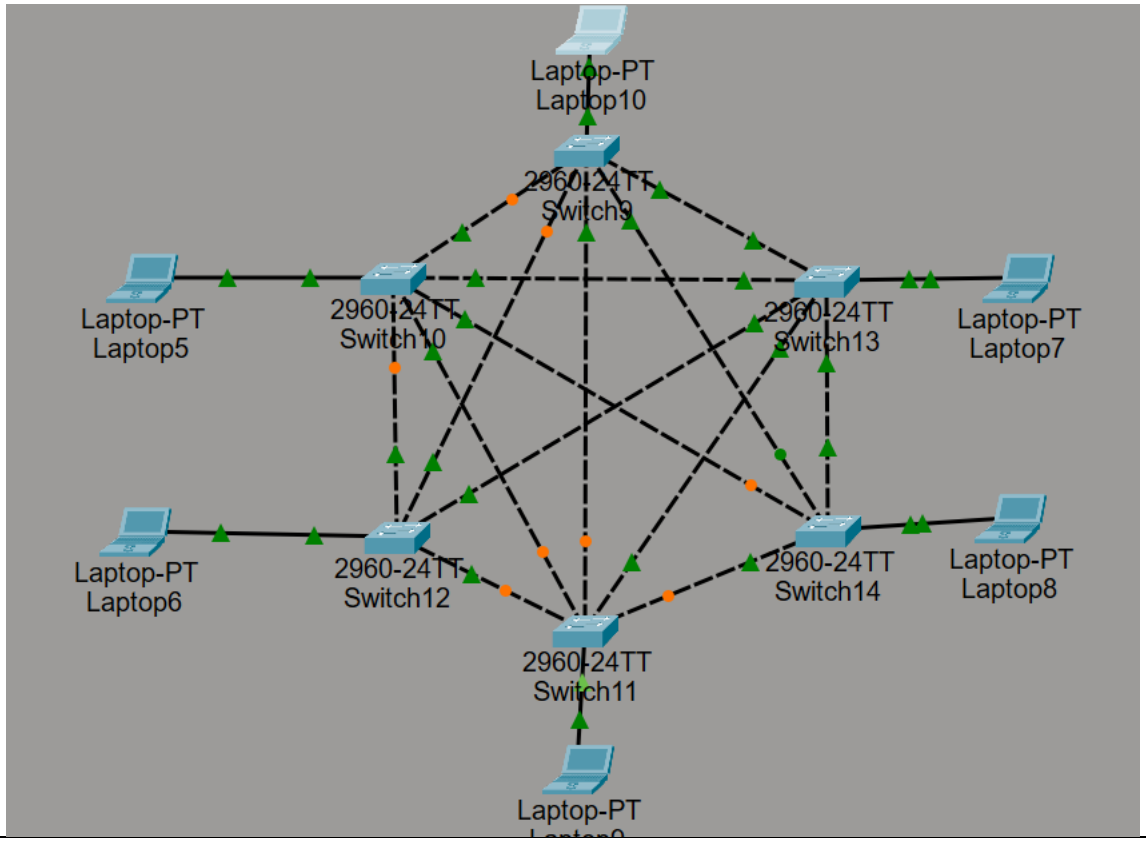
Play Controls

Event List Filters - Visible Events

ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IEC, IPSec, ISAKMP, IoT, IoT TCP, LACP, LLDP, MODBUS, Meraki, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, POP3, PPP, PPPoED, PTP, Profinet, RADIUS, REP, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP

Edit Filters | Show All/None

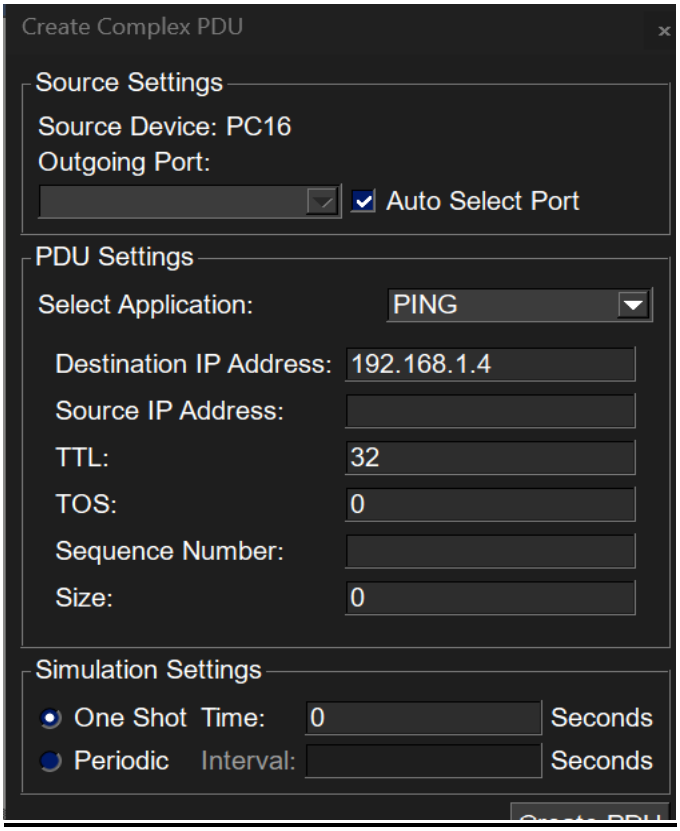
Mesh Topology

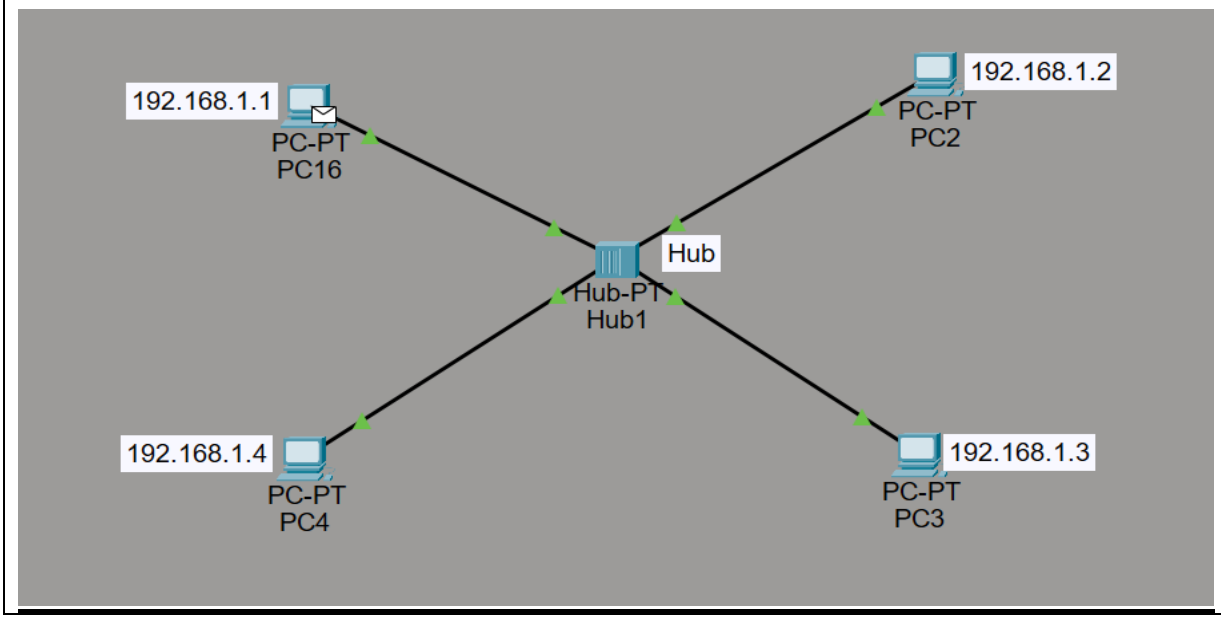


Lab Task 3: Explain the working of Hub using CISCO Packet tracer. Connect PC0,PC1,PC2,PC3 and PC4 to a Hub and send a PDU from PC1 to PC4.

Creating PDU:

We set up this simulation to see exactly how a network hub handles data by assigning unique IP addresses to all the computers and using the custom PDU tool to send a single ping packet directly from **PC1 to PC4**. Because a hub is a basic, unintelligent Layer 1 device, it cannot read destination addresses or route traffic efficiently; instead, it blindly **broadcasts** the packet by flooding copies of the data out to every single connected device on the network simultaneously. When this broadcast wave hits the other computers, **PC2 and PC3** immediately drop the packet because their IP addresses do not match the destination label, while **PC4** accepts it, processes the request, and generates a reply. When PC4 sends its reply back, the hub copies and broadcasts that data to everyone all over again until it finally lands back at PC1, which we tracked step-by-step in **Simulation Mode** to





Conclusion:

In this we understand the network topologies to understand how networking and communication actually work when connecting hubs, switches, and end devices to transfer data from a source to a destination. We also analyzed the actual advantages and disadvantages of each topology, which developed our understanding of speed, efficiency, and security, while also discussing the costs, inefficiencies, and lesser security of certain setups. Furthermore, we learned how to create a PDU and transfer data packets from our source PC to our destination PC in Cisco Packet Tracer using Simulation Mode.